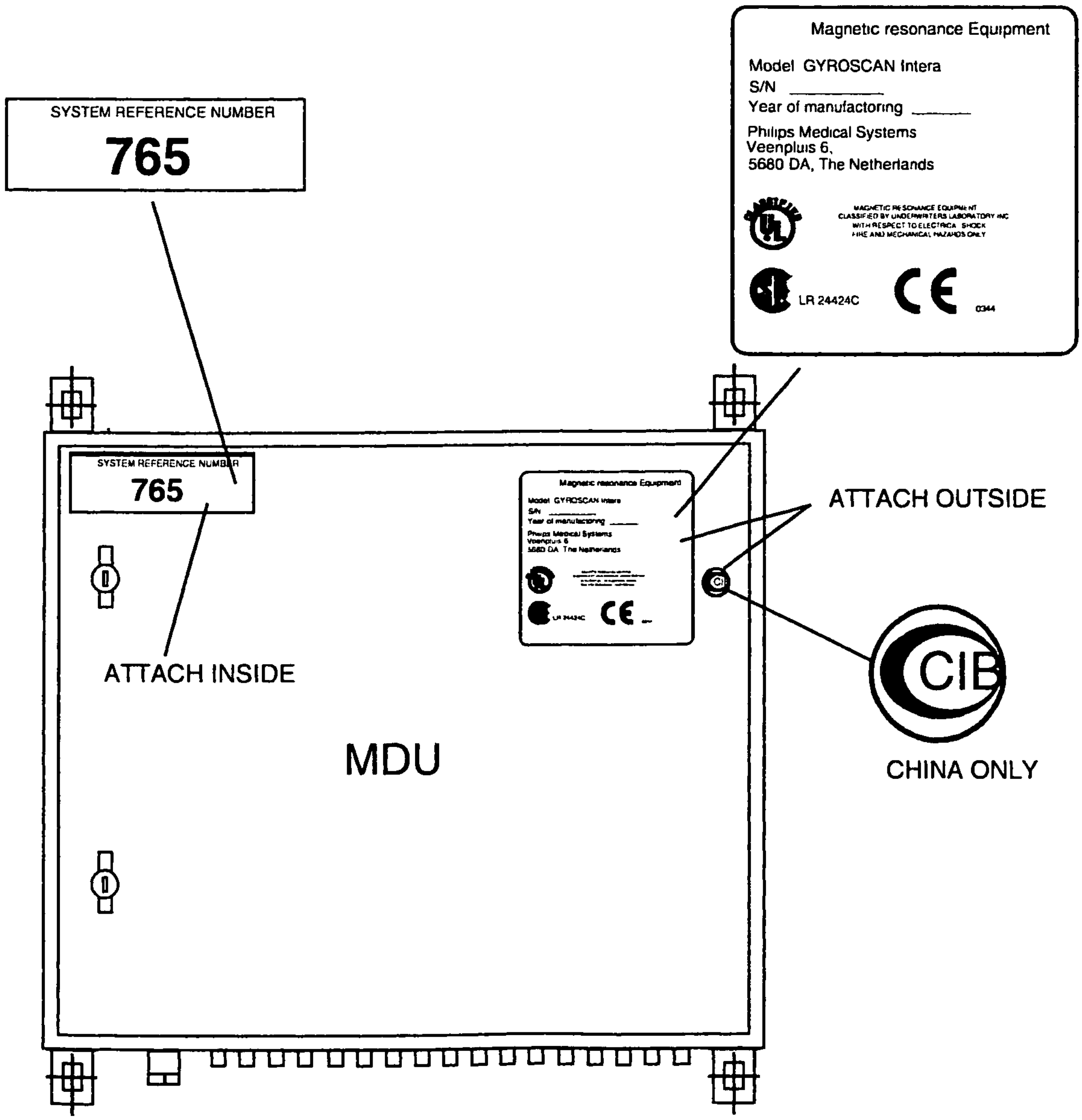
Section 10 Mains Distribution

Contents

1. SYSTEM REFERENCE NUMBER (SRN)

Two Stickers must be attached to the Mains Distribution Unit. The stickers are located in box nr.1 together with the system documentation. One sticker contains the System Reference Number. This System Reference Number is unique for each system and is used for configuration registration, software keys and calling the Helpdesk. The other sticker contains the Gyroscan type, S/N and year of manufacturing. This sticker also contains UL and CE approval numbers for the system as a whole. For inspections of CE or UL it is important that this sticker is attached to the MDU.

Figure 1 – Mains Distribution Unit



2. CHECK INSULATION OF THE FRONT-END

During the installation check for unwanted earth contacts of the magnet and of the patient support.

Insulation between **magnet** and **RF enclosure**

It is assumed that the insulation between magnet and RF enclosure is tested after installation of the magnet as described in Section "Mechanical installation (Part 1)". The value must be written down into the MRL.

Insulation between **Patient Support** and **RF enclosure**

The insulation between Patient Support and RF enclosure must be measured after fixation of the patient support as described in Section "Mechanical installation (part 2)" The value must be written down into the MRL.

Resistance between **Hybrid-box/QBC** and **Gradient coil RF screen (only for Intera Standard and OMNI systems)**:

The insulation between the Hybrid box and the Gradient coil RF screen must be measured after installation of the gradient coil (See section Mechanical Installation (Part 1)). The measured value must be written down into the MRL.

3. EARTH BOUNDING TEST

This test must be performed according IEC 601-1. (current >10 A) e.g. with a WARDRAY bounding tester. The clip should be connected to the protective earth rail in the mains distribution unit and the probe to the metal part of which the ground connection should be measured.

The impedance to the central protective earth terminal of a cabinet may not exceed 50 mOhm.

The impedance to the enclosure of the cabinet or unit may not exceed 100 mOhm.

If the impedance is higher than mentioned above, check the earth cable connections and if needed tighten them.

See MRL for measuring points and specifications.

NOTE

Leakage current measurements must be performed during system level adjustments. This is described in section System Level.

4. CHECK VOLTAGE AT THE MAINS DISTRIBUTION UNIT

Switch "OFF" all units by means of the main switches inside the units and measure the voltage at the following terminals:

For Intera Standard and OMNI systems measure:

Unit	Conn.	MDU switch	Measured between	Voltage
MDU	HX0	Hospital	L1-L2, L2-L3, L1-L3	380 Volt
MDU	HX3	MDU, S1 and F3	N-L1, N-L2, N-L3	220 Volt

For Intera Master and Power systems measure:

Unit	Conn.	MDU switch	Measured between	Voltage
MDU	HX0	MDE	L1-L2, L2-L3, L1-L3	380 Volt
MDU	HX3	MDU, S1 and F3	N-L1, N-L2, N-L3	220 Volt
MDE	HAX0	Hospital	L1-L2, L2-L3, L1-L3	380 Volt
MDE	HAX2	MDE, F2	N-L1, N-L2, N-L3	220 Volt